

2023/7/31 RG presentation

Research theme study

Master 1st year student

Yuki Kusano

Research layer of optical communication

Challenges in optical communication exist at various layers

What challenges currently exist ...

Capture and tracking control technology

Achieve the necessary pointing accuracy to achieve communication

- Acquisition and tracking algorithms
- Vibration mitigation

Communication Technologies

Communication Technologies for Wireless Optical Communication

- Modulation Methods
- Research and Development of Optical Communicators
 - EDFA
 - Antennas, etc.
- Atmospheric Fluctuation Correction

Constellation Technology

How best to configure an optical communication satellite constellation

- Satellite Placement
- Communication paths

This was the topic of my thesis. I hope to go here for my master's thesis as well.

Challenges for capture and tracking control technology

➤ Factors that degrade orientation accuracy

- Alignment errors
- Equipment vibration
 - Reaction wheel disturbances, etc.
- Thermal distortion
- Attitude disturbance
 - SRP
 - Magnetic disturbance, etc.
- Target direction estimation error
 - Sensor accuracy also comes into play
- Control error
 - Sensitivity and minimum output of actuators, etc.
- Sensor error
 - Performance of quadrant sensor, etc.

Errors that are bias-like and almost constant in the short term

- **Alignment error**
- **Thermal distortion**



Since it can be treated almost as a constant value, it is relatively easy to estimate the value and use it to improve the pointing accuracy.

Alignment error and thermal strain

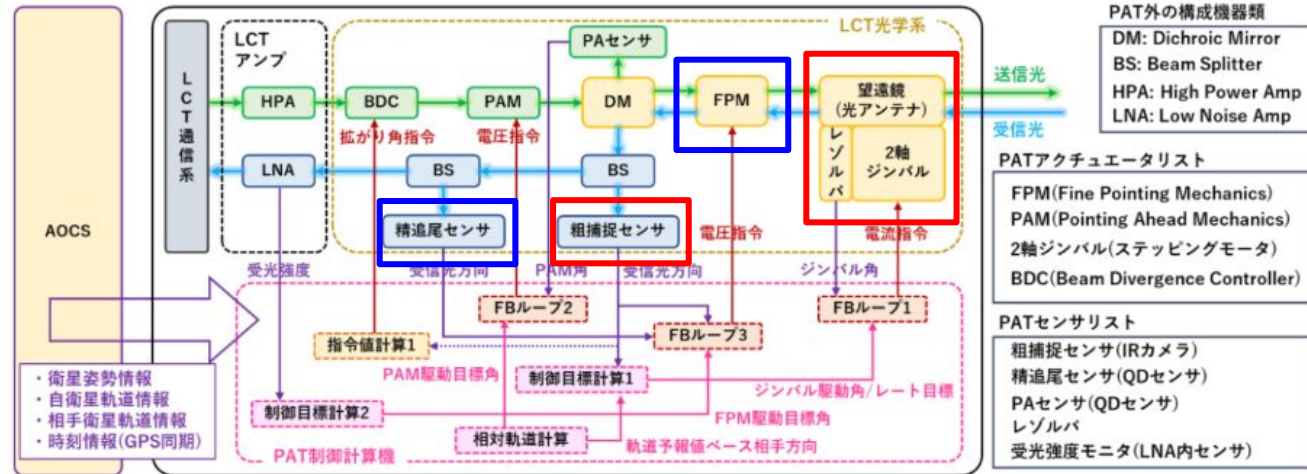
Alignment Errors

- The alignment error described here is mainly due to the assembly process. Therefore, it is not a value that changes dynamically in orbit, but is almost a constant that affects the pointing accuracy in a bias manner.

Thermal Strain

- Earth-orbiting satellites undergo cyclic temperature changes and accompanying thermal distortion due to repeated erosion and non-erosion.
- Powering on and off equipment during communication can also cause temperature changes and thermal distortion due to concentrated heat generation.

Optical Communication System



細沼貴之、平子敬一他、低軌道衛星間光通信ネットワークの実現に向けた捕捉追尾制御アルゴリズムの検討、宇宙科学技術連合講演会、2022

- In many optical communication components, transmitted light and received light follow the same path.
 - In other words, alignment errors and directional errors due to thermal distortion are cancelled out in a fine-tracking system that draws the received light to the center of the sensor.
 - Coarse tracking system not only draws the received light to the center of the coarse tracking sensor, but also performs feed-forward control based on orbit information → alignment error and directional error due to thermal distortion come into play.

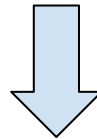
Research Aims

- Establishment of methods for estimating alignment errors and thermal strain in optical communications
 - Examples of high-precision earth observation satellites are useful.
 - It is doubtful that current optical communication satellites have thought much about this area -> there are not many papers, etc,
 - Is the ability to use light as an observable quantity a key point?
- Difficulties in Optical Communication Acquisition and Tracking Systems
 - The factors that degrade pointing accuracy, such as those listed on page 3, are all resolved by feedback control based on a single view of light.
 - The mechanisms of the individual factors themselves are relatively easy to guess physically.
 - For example, there is a theoretical formula for SRP that can be calculated using S2E.
 - However, if you try to guess the actual value of each individual factor, you will fall into the trap of non-observability.
 - Basically, only the directional information (2D) obtained from light exists.

Research Themes

Proposed calibration method using stars

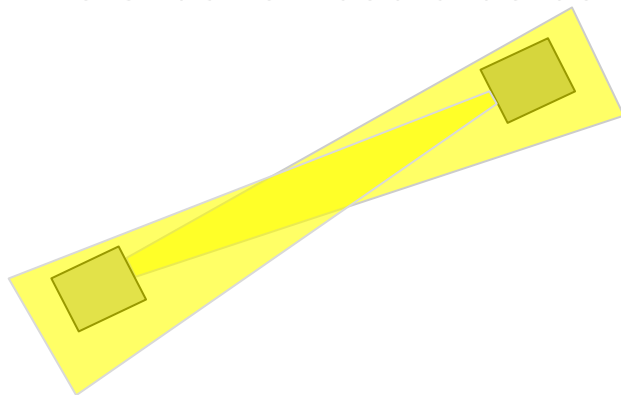
- How to obtain observables for alignment error and thermal strain estimation
- I want to complete calibration before communicating with satellites and HAPS in actual operation.
 - First communication may stumble suddenly, then error estimation is not possible.



Prior to actual operation, the system is pointed and tracked in the direction of a star as a communication partner to estimate alignment errors and thermal distortion. The advantage of this method is that the acquisition and tracking sequence can be checked during the simulated communication with the star.

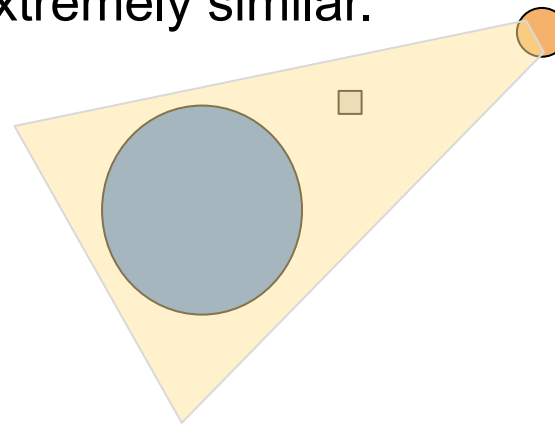
Difference between real communication partners and fixed stars

- The situation could be considered extremely similar.



Inter-satellite optical communication (also for communication with ground stations and HAPS)

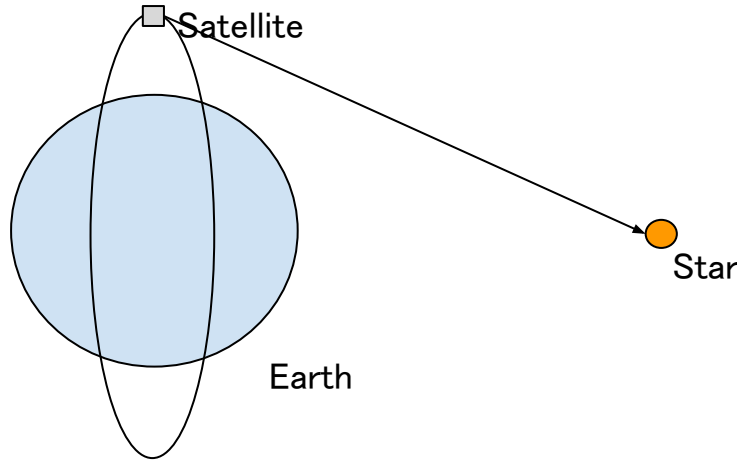
- Communication light completely covers the communication partner in terms of the spread angle of light and the order of communication distance
 - No need for alignment
 - Angle problem



Interstellar Satellites

- Because the star is so far away, the light emitted from it covers not only the satellite but the whole earth.
 - If the satellite itself is pointed firmly in the direction of the star, the light can be observed.

Assumed Observation case



- As mentioned earlier, thermal strain has a rather long periodic cycle due to repeated erosion/non-erosion
 - Observations over several orbital weeks are desired.
- ↓
- The star that is visible at all times during the orbit is considered as a communication partner and is tracked for several orbital weeks.

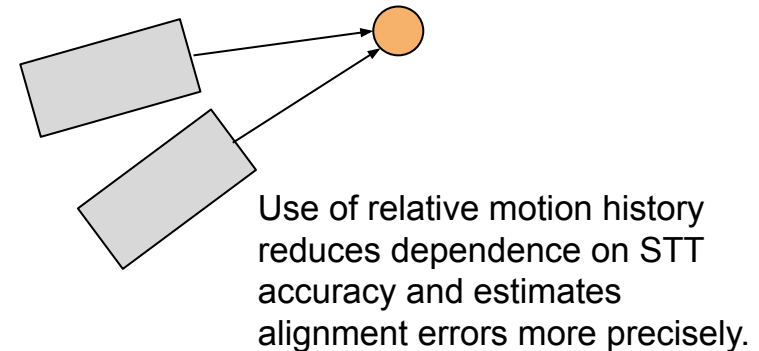
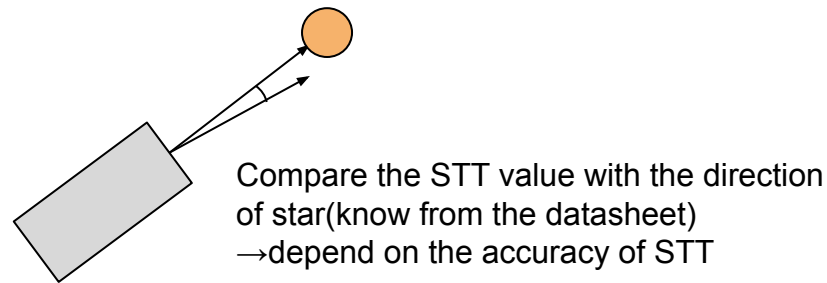
- **Advantages of using stars as calibration partners**

- You already know the position of the star
 - Target pointing accuracy can be determined as long as your position is known
 - No need to estimate the position of the target
- Light from the star is always available.
 - You can receive the light if you can point the star firmly.

We can cut off the element of dependence on the other party.

Estimation

- There are several examples of alignment error and thermal strain estimation using high-precision observation satellites instead of optical communication satellites.
 - Most of them use STTs or image data from onboard optical cameras as the observation data.
- In the case of optical communication satellites, we believe that the use of optical acquisition and tracking mechanisms will enable us to obtain estimation accuracy higher than that of the above-mentioned observation satellites.



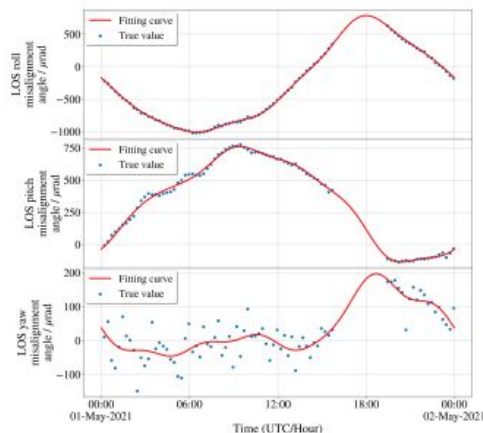
Previous work

- Estimation of thermal strain
 - Hualong Hu, Xialchong Tong, Chunping Qiu, He Li, and et al. On-Board Thermal Motion Compensation Method for Pointing Errors of the Remote Sensor Aboard a Three-Axis Stabilized Geostationary Satellite
 - Paper on Estimation of LOS Directional Change due to Thermal Strain in Geostationary Earth Observation Satellites
 - The effects of thermal strain are caused by a complex combination of various deformations, such as deformation of the lens surface and deformation of the structure itself.
 - In this paper, we do not estimate these effects separately, but rather, we estimate them in the form of three-axis errors, i.e., roll, yaw, and pitch angle errors in the LOS direction.
 - Assuming periodic changes, the errors are modeled in the form of a Fourier series expansion as follows

$$\begin{cases} \varphi_{\text{ma}} = a_0 + \sum_{n=1}^6 [a_n \cos(n\omega t) + b_n \sin(n\omega t)] \\ \theta_{\text{ma}} = c_0 + \sum_{n=1}^6 [c_n \cos(n\omega t) + d_n \sin(n\omega t)] \\ \psi_{\text{ma}} = e_0 + \sum_{n=1}^6 [e_n \cos(n\omega t) + f_n \sin(n\omega t)] \end{cases}$$

Previous work

- Obtain series coefficients from the time-series imaging data of the onboard optical camera



- The estimation is not a real-time estimation, but only a fitting based on a time series of past field data.
- When the next orbit is reached, the previous orbit or the previous estimation is used to take into account the effect of thermal strain.

The result of fitting for the coefficients

Future Policies

To do

- Creating thermal strain and alignment models
 - What is the appropriate way to create a model?
 - Is it sufficient to use only angular error as in previous studies?
- Optimization of calibration method
 - What kind of stellar observation method is appropriate?
 - Observation time
 - What observables should be used?
 - Angular displacement history of FPM
 - Output value history of quadrant sensors
 - What kind of estimation system should be used?
 - It depends on how to make the model.
 - Probably Kalman filter, but EKF or UKF, or even particle filter may be possible (the randomness of the thermal strain may be stronger than expected).
- Real-time estimation
 - After calibration, is it possible to further improve the accuracy of the originally estimated values during communication when the communication actually starts?

Research plan

- Investigating how to model alignment error and thermal strain error.
 - I'm thinking it would be nice if we could do it up to M1.
 - Alignment error may be no problem with mounting error angle.
 - Thermal strain may need to be devised in a way that makes it easier to estimate its value.
- Optimization of calibration method
 - We would like to start with M1 and work on the algorithm in parallel with the above modeling.
 - Check if simulation-based error estimation is enable(perhaps M2?)
 - Now thinking of using S2E-based simulation.
- Real-time estimation (later in the order)
 - If we have a good idea and can connect to real-time error estimation, it may be good to conduct a simulated communication experiment in orbit using Beyond5G experimental equipment.